



Flight into the future: a holistic review of AI-trends, vision, and challenges in drones technology

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Abstract

The use of Artificial Intelligence (AI) and Unmanned Aerial Vehicles (UAVs), also known as drones, is changing the way future communication and networking systems are designed. UAVs can collect data, support wireless networks, and help deliver services from the sky, which makes them an important part of modern technology. To understand these developments, we reviewed almost 250 research papers published between 2015 and 2024. Our review focuses on UAV network design, communication methods, energy management, AI-based optimization, and future challenges. Unlike previous surveys that mainly summarize individual technical domains, this work introduces a new AI-driven UAV classification framework that connects these aspects under one structure. The framework organizes UAV systems across five dimensions—mission adaptability, autonomy level, communication intelligence, scalability, and deployment context—providing a unified way to compare current and future UAV technologies. This analytical structure highlights how artificial intelligence enables UAVs to move from static, pre-defined operations toward dynamic, real-time decision-making and mission-specific adaptation. We found that deep learning and reinforcement learning are the most common AI methods used to improve routing, flight planning, resource use, and network performance. These techniques help UAV networks adapt to changing conditions and reduce communication delays. However, we also found several open challenges, such as improving real-time energy efficiency, increasing security and privacy, managing large drone groups (swarms), and dealing with regulatory and policy issues. By combining this new framework with an extensive literature review, the paper offers a holistic view that not only summarizes past progress but also maps existing gaps and trends for future research. This paper provides a clear summary of current research, explains key trends, and points out gaps such as the need for lightweight AI models and better swarm coordination. The insights from this review can help researchers and engineers build smarter, safer, and more efficient UAV networks in the future.

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Keywords AI-trends · Drones · Network-management · Security-management · UAV · UAV-energy-efficiency

1 Introduction

In recent years, researchers have become increasingly interested in autonomous vehicles. Numerous studies have been published on designing and operating UAV (Unmanned Aerial Vehicle) networks, a concept mainly associated with drone technology driven by artificial intelligence (AI). UAV networks play an essential role in addressing the challenges of rugged terrains, where traditional communication networks struggle or fail. The performance of UAV networks is based on contextual awareness, spectrum efficiency, and topological advantages. Researchers are eager to explore AI to enhance network administration, given the limited human involvement in UAV network operations (Nawaz et al. 2021). The integration of machine-learning methodologies is expanding applications in both academic and commercial endeavors, and recent surveys consolidate the state of the art in AI-enabled UAV networking (Zhou et al. 2024) as well as foundational communication models and design trade-offs (Zeng et al. 2016; Zeng and Zhang 2017). Despite significant progress, most existing studies focus on isolated aspects of UAV systems—such as routing, localization, or flight control without linking them within a comprehensive structure. This fragmentation has created a clear research gap: there is still no unified framework explaining how AI can simultaneously influence UAV autonomy, communication efficiency, energy management, and scalability (Zhou et al. 2024; Zeng et al. 2016; Zeng and Zhang 2017; Rovira-Sugranes et al. 2022).

To address this gap, the present review proposes a new AI-driven UAV classification framework that organizes existing research into a single, interconnected view. The framework is built on five analytical dimensions: (1) *mission adaptability* how UAVs change tasks or roles in real time; (2) *autonomy level* how AI supports decision-making and self-management; (3) *communication intelligence* how networks learn and adapt through AI-driven routing; (4) *scalability* how large UAV swarms coordinate and share information; and (5) *deployment context* how UAVs integrate with specific environments and applications. This view connects AI techniques such as deep reinforcement learning and federated learning to tangible design choices in routing, spectrum use, placement/trajectory, and cooperative control (Zeng et al. 2016; Zeng and Zhang 2017; Rovira-Sugranes et al. 2022; Hentati and Fourati 2020). This multi-dimensional approach enables a holistic analysis of UAV systems from both technological and operational perspectives, linking AI methods with practical design challenges. It also distinguishes traditional static UAV classifications from emerging dynamic, AI-enhanced models that adjust to mission requirements and environmental changes. In addition to introducing this framework, our study contributes by summarizing nearly 250 research papers published between 2015 and 2024 across multiple UAV domains; highlighting major AI algorithms (e.g., deep reinforcement learning, federated learning, graph-based methods) used for UAV optimization (Zhou et al. 2024; Hentati and Fourati 2020; Lahmeri et al. 2021); identifying open challenges in real-time energy management, data privacy, swarm coordination, and lightweight edge AI; and outlining future research directions that connect AI trends with sustainable UAV deployment (Zeng et al. 2016; Rovira-Sugranes et al. 2022; Hentati and Fourati 2020; Lahmeri et al. 2021). Through

Fig. 1 AI-Driven UAV Classification Framework. The proposed framework integrates artificial intelligence (AI) as the central enabler that connects five analytical dimensions of UAV technology: (1) mission adaptability, (2) autonomy level, (3) communication intelligence, (4) scalability and swarm coordination, and (5) deployment context

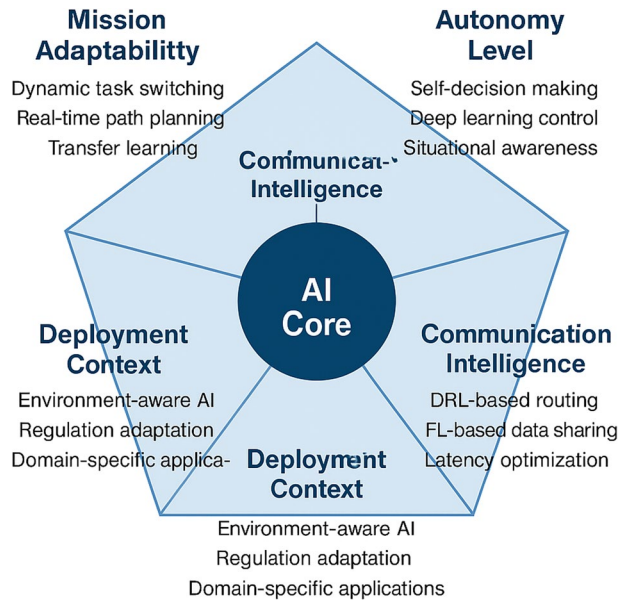


Table 1 Analytical summary of the proposed AI-driven UAV classification framework

Dimension	Focus area	Key AI techniques	Expected outcome
Mission adaptability	Real-time role switching and task reallocation	Reinforcement Learning, Transfer Learning	Flexible mission planning, adaptive re-routing
Autonomy level	Independent perception and decision-making	Deep Learning, Fuzzy Logic, Hybrid AI	Higher operational independence and safety
Communication intelligence	Smart routing and data management	DRL, Federated Learning, GNNs	Reduced latency, improved connectivity
Scalability & swarm coordination	Multi-agent learning and collaboration	Swarm Intelligence, Distributed Learning	Coordinated behavior, large-scale synchronization
Deployment context	Application and environment-specific AI	TinyML, Context-Aware AI, Edge Computing	Efficient operation in domain-specific scenarios

this comprehensive synthesis, the review moves beyond descriptive summaries and provides an analytical foundation for understanding how diverse AI technologies collectively advance the next generation of intelligent UAV networks. Figure 1 shows the taxonomical structure of the AI-based UAV network. Similarity Table 1 is showing the analytical summary of the proposed AI-driven UAV classification framework.

Furthermore, the Fig. 2 shows the taxonomical structure of the AI-based UAV network.

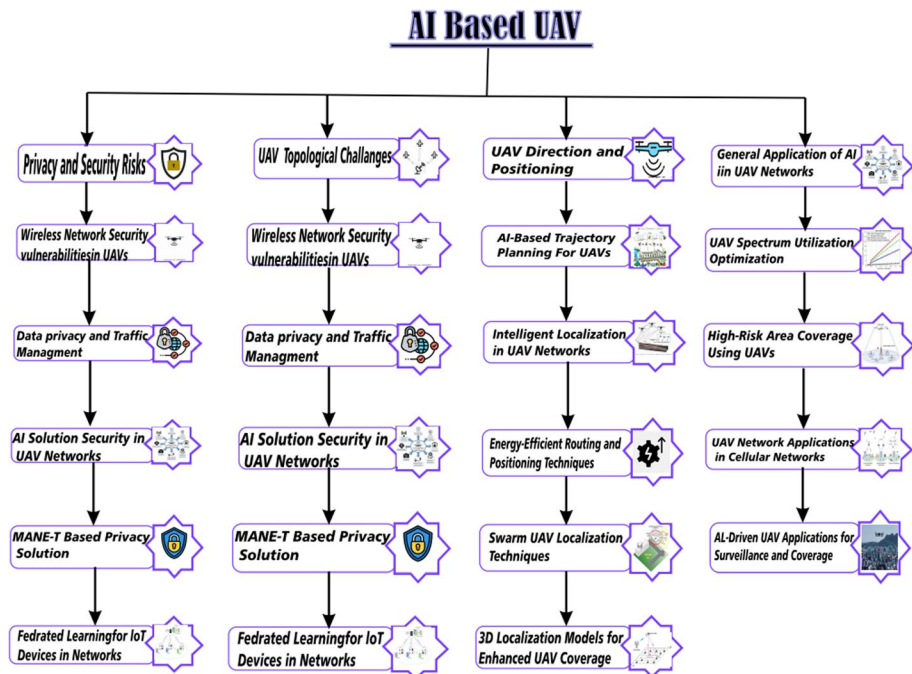


Fig. 2 This diagram illustrates the multifaceted applications of AI in UAV (Unmanned Aerial Vehicle) networks, including privacy, security, and efficient routing. Highlights challenges and solutions within UAV systems such as topological issues, trajectory planning, and spectrum optimization. The flowchart also delves into specific AI-driven security solutions and advancements in UAV localization and coverage techniques

1.1 UAV networks based on AI

AI is advancing and developing strategies that mimic or even surpass human intelligence. The primary strength of AI lies in its ability to learn and adapt (Yao et al. 2024). AI techniques are particularly beneficial for UAV-based networks due to the inherent fluidity and unique challenges of this technology. Researchers are actively exploring artificial intelligence (AI) applications in UAV networks. This section discusses four critical aspects of AI-based UAV network design: location and direction, network architecture, privacy and security, and general UAV applications. These areas are vital for the successful design and deployment of next-generation drone (Letaief et al. 2019).

1.1.1 Privacy and security risks

Two significant risks are associated with UAV networks: the inherent security vulnerabilities of wireless networks and issues concerning data privacy and traffic management in these networks, especially with dynamic topologies. Extensive research has been conducted to address these privacy and security concerns in UAVs. Artificial intelligence techniques have been proposed to address these challenges. For example, in research (Challita et al. 2018), the authors present a solution to address security and privacy concerns using Conventional

Cellular Networks (CCN). However, AI models such as convolutional neural networks (CNN) and recurring neural networks (RNNs) can also address these issues effectively. These models help to classify UAVs according to their movement patterns and identify areas of high risk. A study suggests that nearby UAVs in a swarm exchange mainly localized information (Kusyk et al. 2019). Simulation results indicate that such a UAV swarm can provide broad global exposure by leveraging local knowledge while mitigating cyber threats. Each UAV in the simulation is connected to a Mobile Ad Hoc Network (MANET) via an OPNET Modeler. In addition, a novel approach based on federated learning (FL), a machine learning technique, was developed for distributed devices on the Internet of Things (IoT) (Lim et al. 2021). FL is essential for establishing secure connections between IoT devices. A UAV-based FL model utilizing 5 G heterogeneous network concept, was presented in Dai et al. (2019). The integration of UAVs into FL improves communication efficiency and ensure a more reliable and secure network traffic coverage.

1.1.2 UAV topological challenges

One distinctive feature of UAV-based networks is that methods designed for other types of networks, such as MANET and Vehicular Ad Hoc Networks (VANET), are not directly applicable. Researchers have approached the challenges in UAV networks differently. In Reynolds (1987), an autonomous flock control strategy is proposed to manage the swarm topology of UAVs. The suggested approach maintains UAV formation during flight while minimizing energy consumption. The authors applied the Reynolds-Boid algorithm (Li et al. 2021), which involves the principles of cohesion, separation, and alignment. The proposed method was simulated and analyzed using the OMNET++ Modeler. Furthermore, Li et al. (2021) discusses wireless challenges in UAV networks and offers AI-based solutions. A primary issue in UAV networks is maintaining a stable network with low latency that can efficiently handle dynamic conditions. Several AI model-based methods have been proposed to improve UAV network performance.

1.1.3 UAV direction & positioning

In UAV networks, location and trajectory planning are critical considerations. A recent study proposes AI-based trajectory planning for UAV networks (Arafat and Moh 2019). A quantum mechanism is used to navigate autonomous vehicles (UAVs) from one location to another. According to the study (Fatemidokht et al. 2021), the proposed method outperforms traditional Q-learning approaches. Localization is particularly challenging due to significant node mobility and unstable network conditions. The authors propose an intelligent localization method for swarm UAVs (Chen et al. 2022). A 3D model is used to address the challenge of finding the location of the UAV node, starting by reducing the search space. This approach enhances network convergence performance and reduces localization errors. The proposed method is further strengthened by implementing energy-efficient routing for the UAV network. The simulation results demonstrate that the recommended strategy improves both localization accuracy and coverage time (Gupta et al. 2015).

1.1.4 General applications

Research is ongoing into various UAV network applications, including efficient spectrum utilization, high-risk area coverage, and the use of UAV networks in cellular and vehicular networks. In Khuwaja et al. (2018), the authors discuss the use of UAV networks to enhance VANET performance. The proposed routing techniques improve coverage and detect malicious vehicles. Experimental results based on simulations reveal that approximately 7% of UAVs are identified as high-risk for the UAV network.

Our UAV classification framework extends beyond traditional taxonomies by integrating AI-driven adaptability, real-time decision-making, and deployment-specific optimization. In Table 2, we compare our framework with conventional taxonomies.

By emphasizing real-world UAV deployment strategies, our classification framework offers a more practical and AI-enhanced approach compared to static taxonomies.

1.2 Overview of previous research on UAVs

Recent advancements in autonomous UAV networks have expanded the scope of research covered in survey papers. This section reviews some of the most recent surveys in the field. Coverage is one of the most critical research areas for UAVs. Numerous solutions have been developed to enhance area coverage using UAV networks. A survey (Zhang et al. 2019) identifies the challenges in UAV network coverage, categorizing them into various constraints. The survey findings suggest further research on end-to-end delays, inter-UAV network coverage, and UAV network coverage in general. A 2015 survey (Arafat and Moh 2019) distinguishes UAV networks from other wireless technologies, such as vehicular technologies and ad hoc mesh networks, due to their unique features. The dynamic nature, intermittent connectivity, fluid topology, and energy-conservation limits of autonomous vehicles make UAV networks distinct from other wireless networks. This survey addresses issues in UAV networks at the physical, data link, network, and transport layers. The results highlight that information sharing is necessary to meet the requirements of UAV networks.

The paper (Fotouhi et al. 2019) expands the discussion on UAV network characteristics, focusing on network channel modeling and its impact on performance. A detailed analysis is conducted at lower altitudes to assess the equipped facility. Energy efficiency in UAVs

Table 2 Comparison of traditional UAV taxonomies and the proposed AI-driven classification framework

	Feature	Traditional taxonomies	Our AI-driven classification
The table contrasts static, airframe-based taxonomies with a dynamic AI-driven approach. Our framework focuses on mission adaptability, real-time decision-making, scalability for large UAV swarms, and integration of specific use cases such as VANETs, disaster response, surveillance, and energy-aware missions	Categorization basis	Airframe type (fixed-wing, rotary-wing, hybrid)	Mission-adaptive UAV behavior, autonomy levels, and communication integration
	Adaptability	Static classifications	Dynamic role-switching based on real-time AI-based decision-making
	Scalability	Limited to small UAV groups	Optimized for large UAV swarms in dynamic environments
	Use case integration	Generic UAV applications	Specific AI-powered UAV roles: VANETs, disaster response, surveillance, energy-aware missions

is further examined on air-to-air and air-to-ground surfaces using an empirical modeling approach. A study (Oubbati et al. 2020) explores the use of 5 G technology in UAV networks. The authors propose seven different antenna techniques for UAV networks. Furthermore, a study (Skorobogatov et al. 2020) discusses communication between UAVs using cellular networks. The research demonstrates the successful transition of UAV communication from traditional cellular networks to flying relays and base stations. The study also addresses associated risks and issues regarding security measures, traffic standardization, and privacy concerns, providing valuable insights for future work. The increasing coverage of UAV networks is making them more popular, while industry experts strive to optimize spectrum efficiency.

In 2020, a survey investigated the effective use of NFV and SDN in UAV networks (Zhi et al. 2020). The authors proposed two methods to address inoperability issues in SDN networks and NFV. The paper also discusses areas of weakness in UAV network research and suggests directions for future work. The use of UAV networks in various regions is explored in Bithas et al. (2019), with the aim of classifying UAV networks into nine distinct categories. The paper thoroughly classifies UAV networks and reviews recent research in the field. The study of multiple UAVs identifies three levels of autonomy.

The paper also discusses various challenges associated with UAV networks and suggests how these challenges can be addressed in the future. Numerous researchers, institutions, universities, and scientists are working on new solutions to overcome these challenges and achieve higher accuracy, performance, and efficiency in UAV networks using cutting-edge technologies. A recent survey examines UAV networks from the perspectives of security and privacy. The study identifies key issues that affect the functionality of UAV networks, such as sensor malfunctions, erratic communication between aerial and ground stations, and potential photo leaks from aerial views.

In a study similar to ours, conducted in 2019, Ben Aissa and Ben Letaifa (2021) reviewed several machine learning approaches applied to UAV networks. The authors used ML models in this survey to classify different UAV networks and their components, such as base stations. The paper suggests optimizing UAV network performance by applying machine learning techniques. The researchers propose building a highly efficient UAV network using deep learning techniques and multi-source data. Another study (Ben Aissa and Ben Letaifa 2021) discusses using ML methods for the evaluation, monitoring, observation, and security of UAV networks. UAV networks are particularly suitable for edge computing, where ML models can be easily implemented. The communication challenges between UAVs and base stations can be addressed by increasing base nodes and leveraging mobile edge technology. In Song et al. (2021), a comprehensive study of UAV network communication, prototyping, and experimental setups is provided, detailing the construction of an experimental and prototype UAV network. The study covers every phase of the setup, including aircraft selection and communication techniques, and investigates methods for reliable UAV communication.

1.3 Key contribution of this research

This survey paper addresses the following key research contributions:

1. Develops a classification framework for UAVs based on their features and functionalities to enhance network performance and optimize deployment strategies.

2. Identifies the main challenges facing UAV networks and evaluates the effectiveness of current methods used to address these challenges.
3. Investigates how the integration of artificial intelligence in UAV features influences their overall performance.
4. Analyzes how different channel models (e.g., Shadowing Channel, Free-space Path Loss) impact the efficiency of UAV communication and resource management.
5. Explores the characteristics of UAVs that contribute to or help mitigate operational risks and assesses their impact on overall mission effectiveness.
6. Investigates emerging security threats to UAVs and determines the need to develop new features to enhance UAV security.
7. Assesses how variations in payload capacity and flight range impact UAV performance across different mission durations and deployment environments.
8. Formulates strategies to extend the communication capabilities of UAVs, particularly for long-range missions in remote areas.
9. Explores methods to integrate various sensor capabilities (e.g., for surveillance, delivery, mapping, inspection) within UAVs to maximize mission flexibility and adaptability in diverse deployment scenarios.
10. Identifies advancements needed to enhance the autonomy of hybrid drone designs for improved performance and maneuverability in real-world operations.

The first section introduces AI-based UAV networks and provides a summary of previous research, highlighting the most significant contributions in the field. Section II presents background information and preliminary data from earlier published work on the subject. Section III delves into the self-governance of UAV networks, network connectivity, base stations, cutting-edge technologies, and other relevant UAV aspects. Section IV covers UAV features, resource management, network architecture and planning, access points, and various communication protocols, including energy efficiency and management strategies. Section V addresses UAV-related privacy and physical layer security risks, with a focus on identifying research challenges. Finally, Section VI concludes the research and discusses future directions for work in this domain.

2 Literature review

UAV networks present a novel communication method distinct from traditional approaches. This section provides an overview of the background and fundamental concepts that were explored during the early development of UAV networks.

2.1 Conventional vs. unmanned aerial vehicle networks

The global drone market is projected to reach a value of USD 70.5 billion by 2025, according to Business Insider (Insider 2021). UAVs have existed since 1917, when they were first used during World War I (Valavanis 2018). Initially unreliable, UAV technology has advanced significantly, with improvements in accuracy, privacy protection, and communication capabilities. Recent research breakthroughs have led to more robust UAV network solutions. UAV applications are expanding across telecommunications, security, surveillance, and mil-



Fig. 3 The image shows a fixed-wing UAV (Unmanned Aerial Vehicle) with labeled arrows pointing to its fixed wings. The UAV is grounded on a flat surface under a clear blue sky. The caption underneath the image reads "(a) Fixed wing"



Fig. 4 The image shows a multi-rotor UAV (Unmanned Aerial Vehicle) with labeled arrows pointing to its rotors. The UAV is hovering in the air, supported by multiple rotors attached to its frame. The caption underneath the image reads "(b) Multi-rotors"

itary operations. Due to their adaptability and cost-effectiveness, UAV networks are ideal for research and communication in challenging environments such as forests, mountainous regions, and riverbanks. When considering UAV design, aerodynamics plays a crucial role in determining their lifting mechanisms. UAVs are typically classified as fixed-wing (Fig. 3) or multirotor (Fig. 4), depending on their lifting methods (Garg 2022). Multirotor UAVs generate lift using vertical thrust, often employing two to eight motors. These UAVs are equipped with vertical takeoff and landing technologies, although they consume significant energy during operation. In contrast, fixed-wing UAVs, which rely on gliding to save energy, offer greater efficiency. Ongoing research on fixed-wing UAVs focuses on enhancing their operational capabilities (Panagiotou et al. 2016). Unlike traditional aircraft, fixed-wing UAVs require runways for takeoff and landing. In UAV networks, both fixed-wing and multirotor UAVs are commonly used. Fixed-wing UAVs are suitable for long-range mis-

sions due to their efficiency, while multirotor UAVs are preferred for tasks requiring precise maneuverability and vertical takeoff.

2.2 Communication, computation, and control

UAVs have the potential to be highly beneficial in many aspects of daily life. Cooperative UAVs represent an emerging concept that aims to enhance the capabilities of individual UAVs through collaboration. Numerous research studies are currently exploring the potential of such UAV networks. For example, Richards and How (2004) proposes a decentralized approach for predictive maintenance, using a fleet of UAVs to demonstrate the model. The simulation results show that the proposed model is more computationally efficient than other approaches. Bellingham et al. (2003) conducted an intriguing study focused on developing a control system that optimizes multi-UAV collaboration. The key issues addressed include task assignment, resource allocation, and trajectory optimization. The proposed approach improves UAV fleet management by incorporating cooperative route planning and multitasking through the integration of trajectory and resource management. When multiple elements work together, conflicts may arise. A study on UAV fleet conflict resolution is presented in Conde et al. (2012), proposing a model to identify and resolve conflicts. The authors introduce intermediate waypoints to structure flight paths and maintain consistent speeds for UAVs. Experimental and simulation results demonstrate that the proposed system is scalable and easy to implement. The study also suggests utilizing thermal lift to extend the flight time of cooperating UAVs.

Traditional UAV target-tracking methods rely on predefined flight paths, limiting their adaptability in dynamic environments. Deep Reinforcement Learning (DRL) enables UAVs to learn real-time trajectory adjustments using sensor feedback (Debnath et al. 2024), allowing them to optimize flight paths autonomously. A Multi-Agent Reinforcement Learning (MARL) framework was introduced for UAV swarm-based surveillance (Ma et al. 2023), demonstrating a 35% increase in tracking accuracy compared to heuristic-based approaches by allowing UAVs to dynamically adjust routes in response to moving targets and obstacles (Patrizi et al. 2020). DRL-based models excel in handling real-time occlusion scenarios, where UAVs dynamically modify observation angles to regain lost targets, proving highly effective in security surveillance, search-and-rescue operations, and disaster response scenarios (Katkuri et al. 2024).

UAV swarm coordination faces computational bottlenecks when tracking multiple targets in large-scale deployments (Zhou et al. 2021). Graph Neural Networks (GNNs) enhance UAV intelligence by enabling spatiotemporal data sharing, optimizing multi-target tracking and obstacle avoidance. A study (Phadke et al. 2023) proposed a graph-based trajectory optimization model, achieving 28% faster target reacquisition than traditional A* search-based approaches by leveraging dynamic inter-UAV communication and adaptive flight path planning (Pan et al. 2023). GNNs allow UAV swarms to operate efficiently without centralized control, making them ideal for military reconnaissance, real-time environmental monitoring, and disaster relief operations (Qu et al. 2021).

Multi-agent learning approaches have shown strong potential for large-scale UAV coordination, enabling real-time task allocation and collision avoidance. Compared to centralized coordination models, decentralized AI-based systems improve scalability and fault tolerance. However, current swarm models still lack stability in dynamic environments, and

training multi-agent systems remains computationally expensive. Lightweight cooperative learning frameworks could address these limitations by combining local decision autonomy with shared global optimization. Federated Learning (FL) is revolutionizing UAV network intelligence by enabling decentralized model training, reducing bandwidth usage, and preserving data privacy (Li et al. 2024). Unlike centralized AI models that require continuous data transmission, FL allows UAVs to train locally and share model updates without exposing raw data. One study implemented FL in UAV-assisted mobile edge computing, achieving 40% energy savings and reducing tracking latency compared to traditional cloud-based AI processing. This method enhances real-time adaptability in smart city surveillance, autonomous vehicle monitoring, and emergency response networks, where UAVs can collaboratively learn mission-specific patterns while maintaining privacy and security (Ahmad et al. 2024).

2.3 Channel modeling

Channel modeling is a critical aspect of UAV networks. As UAVs are expected to become increasingly prevalent in the aerospace sector, discussions about channel usage are essential. In 2019, a comprehensive investigation into UAV channel modeling was conducted (Yan et al. 2019) (Nimma and Zhou 2024). The survey categorizes the usage of the UAV channel into air-to-ground, air-to-air, and ground-to-ground communication. The authors also explore channel fading characteristics and link budget considerations. The temporal and spatial properties of non-stationary channels are further analyzed with a particular focus on scientific challenges such as shadowing of the plane structure. In Bithas et al. (2020), a new channel model is proposed to enhance the performance of UAV communication under shadowing conditions. The goal is to establish a generalized channel model with fewer configuration options. The concept and methodology are supported by empirical data.

2.4 Handling of interference

Interference management is crucial in any wireless communication system, and it plays a significant role in UAV communication performance. Recent studies have explored interference management for UAV networks, either as an extension of cellular networks or in conjunction with power regulation techniques. For example, Challita et al. (2019) analyzes a path-planning system for UAV networks that focuses on interference and latency in terrestrial networks. The study examines interference mitigation using game theory and proposes a deep reinforcement learning system. Each UAV selects its path to minimize delays and reduce interference with the terrestrial network. A similar approach is discussed in Challita et al. (2018), which employs game theory to optimize the energy efficiency of UAVs while minimizing latency and interference with the terrestrial network. In Fouda et al. (2019), UAVs are used to support cellular networks, focusing on interference reduction at multiple levels to enhance IAB (In-Band Backhaul) cellular networks. The study considers various factors and aims to optimize overall network performance, surpassing previous methods.

Effective interference management also requires power control. In Wang et al. (2019), machine learning is suggested as a means of reducing interference by controlling UAV power levels and positioning. The proposed system uses affinity propagation and K-means clustering to minimize overall network disturbances. Simulation results demonstrate the

potential for efficient interference regulation. In Zhang et al. (2019), the issue is tackled from a different angle by splitting the problem into two sub-problems: determining the optimal trajectory for a given transmit power and finding the optimal transmit power for a specific UAV trajectory to control interference. This approach helps identify the best overall solution for the network. Another study (Zhang et al. 2020) addresses the issue by recommending UAV clustering combined with power control to reduce interference. The solution is presented through a game-theoretical approach designed to optimize the overall data rate of the system.

Recent studies on security and privacy in AI-based UAVs explore CNNs, RNNs, and federated learning (FL) for enhanced data protection in UAV-IoT networks, improving communication efficiency in 5 G environments, although scalability and latency remain challenges (Chen et al. 2021; Sanchez-Lopez et al. 2019; Primatesta et al. 2019). UAV network topology and flock control leverage Reynolds-Boid algorithms and quantum-based trajectory planning to achieve energy-efficient formation control and accurate localization, but these approaches demand high computational resources (Fonseca et al. 2021; Yin et al. 2019; Wu and Zhang 2017).

AI-based communication and channel modeling focus on A2G (air-to-ground) and A2A (air-to-air) links, integrating 5 G for interference management; however, applications remain largely limited to low-altitude UAVs (Pandey et al. 2019; Lupascu et al. 2019; Wu et al. 2021, 2019; Cai et al. 2018). Cooperative UAV fleet management enhances task allocation, trajectory optimization, and predictive maintenance, but multi-UAV coordination remains computationally intensive (Li et al. 2018; Sun et al. 2018; Zeng et al. 2020). Interference and power control techniques apply game theory, deep reinforcement learning, and clustering to improve energy efficiency and latency, although extensive modeling is required (Min et al. 2018; Yue et al. 2018; Amphawan et al. 2021; Yang et al. 2019; Konstantinou et al. 2017; Lyu et al. 2016).

Quantum navigation methods optimize UAV trajectories with higher precision, outperforming classical models but requiring specialized computation (Ho et al. 2010a). AI-enhanced VANET-UAV integration strengthens routing security and coverage, although compatibility with existing vehicular networks remains a concern (Ho et al. 2010b). UAV coverage studies provide insights into inter-UAV network efficiency, though many lack technical depth (Ho et al. 2011). Comparisons between UAV and traditional networks highlight UAV-specific strengths but often overlook hybrid networking possibilities (Xue et al. 2019).

NFV and SDN applications in UAV networks enhance interoperability and scalability, but their complex deployment remains a limitation (Zhou et al. 2018). UAV classification frameworks segment UAVs into nine autonomy levels, offering structured guidelines but lacking real-world implementation examples (Jiang and Swindlehurst 2012). Machine learning (ML) models improve UAV classification and optimization, yet practical deployment remains underexplored (Xiao et al. 2016). ML techniques also enhance UAV network monitoring and security, improving edge computing performance, although most findings are simulation-based (Xie et al. 2014).

Studies on UAV communication and prototyping provide insights into experimental UAV wireless setups, but high costs and scalability challenges persist (Chen et al. 2018). Market reports such as Business Insider's UAV industry analysis highlight commercial adoption trends but lack technical depth (Ren et al. 2019). Historical reviews of UAV development contextualize technological advancements but primarily focus on legacy UAVs (Tan et al.

2019). Aerodynamic studies of fixed wing UAVs explore energy-efficient gliding strategies to optimize long-range missions, but hybrid UAV designs require further research (Liu et al. 2019; Abbasi et al. 2020).

3 Research methodology

The literature review section of this article is based on a comprehensive selection of research papers, including surveys, review articles, and original research studies, as shown in Figs. 5 and 6. These articles were carefully chosen from various high-impact journals.

To collect relevant research on AI-based UAV networks, we followed a structured review process. We searched major academic databases, including IEEE Xplore, SpringerLink, Scopus, and Google Scholar. The keywords used in the search included “UAV communication,” “drone networks,” “artificial intelligence,” “machine learning,” “deep learning,” “energy management,” and “swarm coordination.” The search covered research papers published between 2015 and 2025, which is the period that includes the most important developments in this field. We applied clear inclusion and exclusion criteria to select papers. We included peer-reviewed journal articles and conference papers written in English that focus on UAV network architecture, communication methods, energy efficiency, and AI-based optimization. We excluded papers that were not in English, studies unrelated to UAV networking and AI, patents, and papers that focused only on drone hardware without communication or AI aspects. The selection process was done in two stages. First, we screened titles and abstracts to remove duplicates and irrelevant results. Second, we reviewed the full text of the remaining papers to confirm their relevance. After this process, a total of 245 research papers were selected for detailed study. These papers were then grouped into key themes such as communication protocols, energy management, AI optimization techniques, swarm coordination, and security and regulatory issues, which form the structure of this review. The entire process is also explained using flow diagram as shown in Fig. 7.

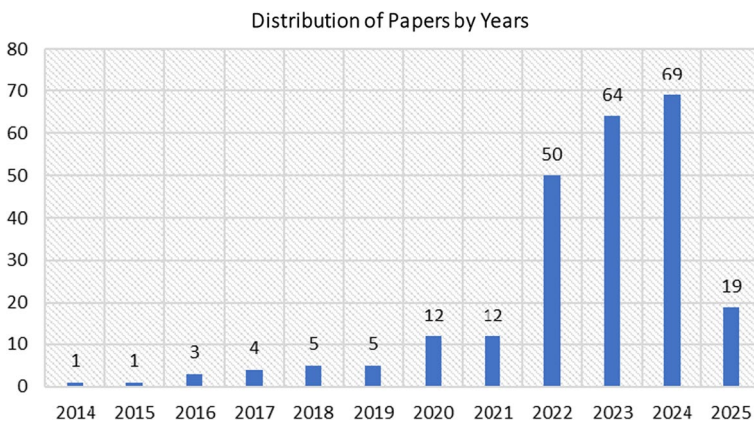


Fig. 5 Yearly distribution of selected research papers (2018–2024). The bar chart illustrates the number of relevant papers published each year from 2018 to 2024. Research activity peaked in 2019 with 20 papers, followed by consistent contributions between 2020 and 2022. A decline is observed in 2023 and 2024, reflecting either a narrowing of the selection criteria or a shift in publication trends

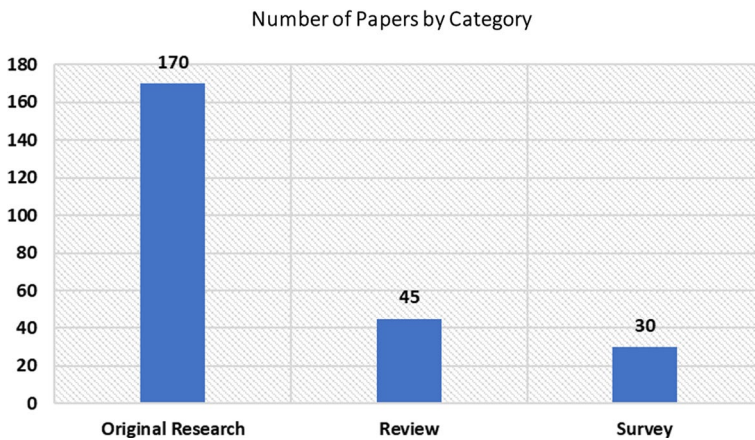


Fig. 6 The bar chart shows the distribution of research papers by journal category. IEEE has the highest number with 33 papers, followed by Science Direct with 19 and Springer with 18. Cornell University has the fewest, with only 6 papers published

4 Autonomous features in UAV networks

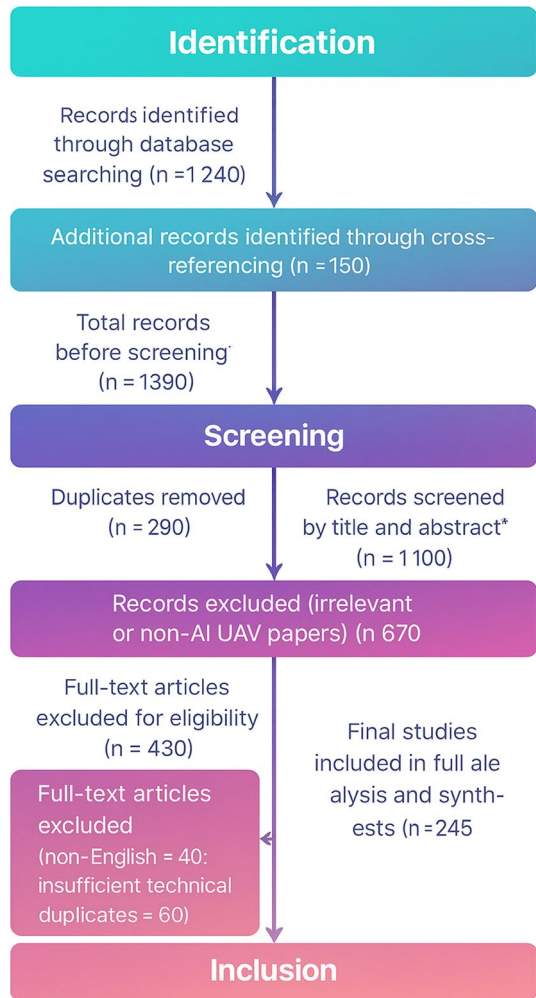
New autonomous capabilities are being developed to enhance UAV network performance and address specific challenges. These approaches aim to provide optimal solutions for particular issues while supporting dynamic network environments. Numerous studies have examined the characteristics of autonomous UAV network management techniques. Figure 8 outlines three different resource management strategies for UAV networks. These strategies include traffic control, transmission control protocols, communication protocols, and efficient power management.

4.1 Network planning and management of resources

Planning and resource management are critical for the success of any network, especially those that involve minimal human interaction, such as unmanned aerial vehicle (UAV) networks. In the context of UAVs, numerous studies focus on addressing the challenges and risks inherent in these networks. These studies aim to overcome obstacles and maximize the potential of UAV networks. Research in this field highlights several key challenges, including dynamic environments, communication constraints, energy limitations, and safety and security concerns. The published work covers a wide range of topics, such as path planning algorithms, resource allocation strategies, risk assessment and mitigation, energy-efficient routing protocols, collaborative task allocation, autonomous navigation systems, and privacy-preserving data collection, as summarized in Table 3.

Research by Nasir et al. (2019) discusses autonomous resource allocation and management in unmanned aerial vehicle (UAV) networks. Table 3 provides a comparative overview of published research on UAV-related topics, focusing on key features such as route planning, conflict resolution, and energy management. It categorizes the studies based on their use of AI, channel modeling approaches, and security considerations. The analysis reveals that while many studies incorporate AI for UAV planning, few address security risks. The

Fig. 7 The flowchart illustrates a process for finding and downloading research papers on web-phishing detection. It starts with a keyword search, followed by checking whether relevant query results are found. If results are found, the user filters the query results and selects more specific papers



diversity in channel models highlights the variety of approaches adopted depending on the research context, making this table highlight a valuable summary of existing work in UAV research. Using game theory, the authors propose several methods for managing UAV resources, including coalition, prospect, pictorial, mean-field, and Stackelberg models. The paper explains the objectives, arrangements, and strategies of each model, highlighting their potential application domains. Another key study (Sun et al. 2018) focuses on real-time UAV route planning in dynamic environments. The authors propose a discrete approach combined with a probabilistic network to create collision-free paths. Similarly, (Nguyen and Le 2019) investigates UAV route planning by integrating data from both static and dynamic routes, proposing a systematic adaptive path planning method to achieve optimal results. An intriguing concept explored in Cui et al. (2019) considers UAVs as additional users in 5 G cellular networks, addressing the unique challenges faced by service providers when integrating these new aerial users.

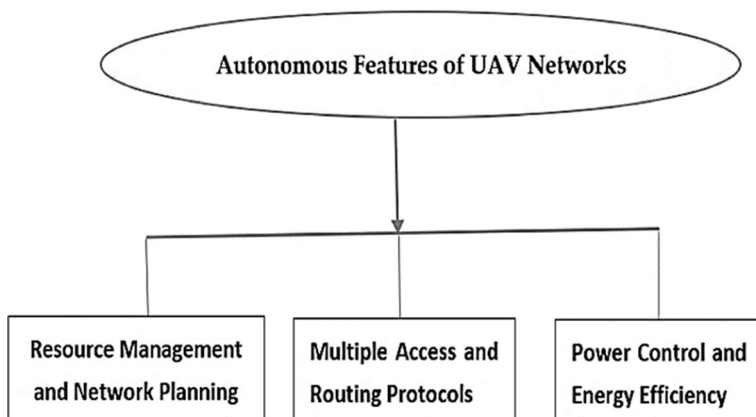


Fig. 8 The diagram highlights the key autonomous features of UAV networks, organized into three main categories: resource management and network planning, multiple access and routing protocols, and power control and energy efficiency. These features are connected under the broader concept of "Autonomous Features of UAV Networks."

4.1.1 Critical insight (Table 3)

Table 3 shows useful methods but also makes the trade offs clear. Older path planning and MAC (Medium Access Control) designs are simple, stable, and use little energy, so they work well in networks that do not change much. However, they cannot adjust quickly when the network changes often, such as in busy urban areas or during fast UAV movements. On the other hand, AI-based methods can adapt in real time and make smarter decisions for routing and communication, which helps in complex and changing environments. But these methods need more energy, computing power, and sometimes special hardware, which can reduce flight time and make drones more expensive. Another important issue is that many studies do not focus on security and privacy. This can leave the network open to attacks like jamming, spoofing, or data theft, which is risky for missions involving sensitive information. Overall, Table 2 shows that there is still no single solution that combines quick adaptability, low energy use, and strong security. Future work should try to design methods that balance these factors, for example by mixing traditional stable methods with AI approaches to get the benefits of both.

4.2 Multiple access and routing protocols

Managing consistency, accuracy, and transmission in UAV networks with multiple access points is challenging. Several studies have focused on addressing these issues. Below is a breakdown of the research findings:

4.2.1 Dynamic channel access

The study focuses on cyclic multi-channel access using computational intelligence for enhanced performance. It addresses line-of-sight (LOS) communication in scenarios involv-

Table 3 Comparative overview of existing UAV research highlighting key features, AI integration, channel modeling, and security considerations

Paper	List of UAV features	AI-Based/ Not AI-Based	Model of channel	Is risk/not	Is se- cure/not secure
Collaborative Route Design and Collisions (Richards and How 2004)	Planning of route	✓	Non-convex modeling	–	–
Planning a Joint Course and AI-locating Resources (Bellingham et al. 2003)	Planning of route	✓	–	–	–
UAV Collaboration and Trajectory Conflicts (Conde et al. 2012)	Conflict detection and resolution	✓	–	–	–
Modelling of Channels (Bithas et al. 2020)	–	✓	Shadowing channel	–	–
Management of Interference (Challita et al. 2019)	Planning of route	✓	Rician distribution	✓	–
Management of Resources (Chen et al. 2021)	Energy consumption, transmission power	✓	Various	✓	–
Navigation Without Collisions (Sanchez-Lopez et al. 2019)	Trajectory planning	✓	–	–	–
Risk-Concerned Route Design (Primatesta et al. 2019)	Planning of route	✓	–	–	–
Management of Resource (Yin et al. 2019)	User defined	✓	Free-space-path-loss	✓	–
UAV Communication security (Min et al. 2018)	UAV-defense	✓	–	–	✓
Ratio of Software-Defined (Yue et al. 2018)	Localization of unwelcome UAVs	✓	–	✓	✓

The table summarizes the primary focus of each study, such as route planning, interference management, and UAV defense, while indicating whether AI-based methods are employed. It also outlines the type of channel models used (e.g., non-convex, shadowing, Rician) and whether the studies address risk and security aspects, providing a clear snapshot of methodological diversity and research gaps in UAV network management

ing the loss of free-space paths but does not cover interference management or security concerns (Sohail et al. 2018).

4.2.2 MAC protocol for UAV networks

This research examines energy-efficient operations and packet error rate monitoring through computational intelligence techniques. It considers free-space path loss with LOS communication but omits discussions on interference management and security (Liu et al. 2019; Jaafar et al. 2020).

4.2.3 Evaluating MAC performance in UAV networks

The study emphasizes packet error rate estimation while evaluating MAC protocols in UAV networks. It utilizes computational intelligence techniques and incorporates Rician fading for channel modeling. However, it does not explicitly address interference management or security concerns (Joudeh and Clerckx 2017).

4.2.4 Trajectory optimization strategies

This research focuses on optimal trajectory planning using computational intelligence for performance enhancement. It considers various channel models such as the Rician K factor, Rayleigh fading, correlated Rician fading, and free-space path loss with LOS. Although interference management is discussed, security aspects are not Mao et al. (2019).

4.2.5 Wave UAV cellular network design

The study explores beamforming strategies for Wave UAV cellular network design using computational intelligence techniques. It considers quasistatic fading and Rayleigh fading for channel modeling but does not address management risks (Xu et al. 2020).

4.2.6 Time-modulated array for channel access

This research focuses on performance measurement and beamforming techniques using time modulation for every access channel. The channel modeling method explains how channels are modeled using LOS and free-space paths with computational intelligence. However, interference risks are not discussed (Yin and Clerckx 2020).

4.2.7 Advanced trajectory optimization

This paper discusses power control combined with trajectory planning, energy management, and computational cost considerations. It uses AWGN (Additive White Gaussian Noise) for channel modeling but lacks interference management methods and security measures (Bansal et al. 2022).

4.2.8 MAC protocol with power optimization

This paper discusses energy management using computational techniques. It handles Rician fading for channel management but does not address security mechanisms (Zhai 2018).

4.2.9 Optimizing throughput in MAC protocols

To enhance UAV network throughput, this paper presents a mechanism to improve performance using MAC protocols and computational intelligence. The author introduces the Free-Space Path Loss method for channel modeling but, like others, does not discuss security concerns in UAV networks (Zhang et al. 2018).

4.2.10 MAC protocol with power optimization

The present research primarily investigates power efficiency in line-of-sight (LOS) communication, employing computational intelligence techniques to enhance overall system performance. It addresses challenges associated with free-space path loss and interference management in LOS environments; however, it does not incorporate or evaluate security aspects (Dutczak 2018).

4.2.11 Adaptive MAC protocol for UAVs

This research provides an overview of trajectory planning, resource management, and energy efficiency using the MAC protocol for UAVs. The author discusses how UAVs communicate using LOS and NLOS models through computational methods. However, security measures for traffic management and communication are not covered (Cho et al. 2021).

4.2.12 Power-aware MAC protocol for UAV networks

Several publications focus on power-aware MAC protocols for UAV networks. This paper discusses how UAV networks efficiently manage power within the MAC protocol. It highlights the use of LOS and NLOS communication channels for energy and power management in smooth transmissions (Geertjes and Spijksstra 2020).

4.2.13 Integrated power and placement optimization

This research addresses optimal power and placement optimization using computational intelligence techniques. It considers channel modeling with AWGN and interference management but does not discuss security aspects (El-Atab et al. 2021).

4.2.14 Rate-splitting techniques

The study explores adaptive beamforming and rate-splitting techniques using computational intelligence. It considers channel modeling with AWGN but does not address interference management or security concerns (Suewatanakul et al. 2022).

4.2.15 Enhancing spectral efficiency in MAC

This research aims to enhance spectral efficiency in MAC protocols through advanced computational intelligence techniques. It considers channel modeling with AWGN but does not address interference management or security aspects (Boukoberine et al. 2021).

4.2.16 Addressing AI algorithm complexity

AI-powered UAV networks often require high computational resources for real-time decision-making, raising concerns about energy consumption, processing latency, and deployment feasibility (Shehzad et al. 2020). To address this issue, TinyML (Tiny Machine Learning) (Kiani et al. 2021) offers a promising solution by enabling AI models to run efficiently on low-power microcontrollers. Techniques such as quantization, pruning, and knowledge distillation can significantly reduce AI model size and computation demands without sacrificing accuracy. In UAV networks, interference is a significant challenge for seamless communication. Several risks, including security breaches and transmission breakdowns, are associated with these networks. Effective interference management is crucial and can be achieved using independent orthogonal multiple access methods for continuous communications. This method is well explained in Cabreira et al. (2019) and Shivgan and Dong (2020). Another study recommends Time Division Multiple Access (TDMA) and Frequency Division Multiple Access (FDMA) to address interference issues. Some studies advocate using Code Division Multiple Access (CDMA) to counteract interference (Peng et al. 2019; Mukherjee et al. 2014). Further research explores Orthogonal Frequency Division Multiple Access (OFDMA) (Huang and Swindlehurst 2011; Zhou et al. 2018; Li et al. 2019) and Space Division Multiple Access (SDMA) (Chang et al. 2017; Wang et al. 2016; Sella-Villa 2020; Wheeb et al. 2021; Dogru and Marques 2022; Nimma and Zhou 2024). However, the limited availability of orthogonal communication channels may reduce spectrum efficiency. Considerable efforts are being made to overcome this challenge by applying non-orthogonal techniques, such as joint optimization of base stations (BS) considering factors such as power (Guillen-Perez and Cano 2018; Guillen-Perez et al. 2021; Kandeel et al. 2022), trajectory (Banafaa et al. 2024; Fotouhi et al. 2019; Lim et al. 2022), and positioning (Yuniarti 2018; Wu et al. 2024).

Non-orthogonal multiple access techniques can perform better, especially when serving multiple users over correlated yet non-orthogonal channels. However, these solutions may fail in scenarios with more antennas than users. Research is ongoing to find alternative solutions to address these challenges. One potential solution is using rate-splitting multiple access algorithms, which have been the subject of extensive research. Much of the field's research focuses on optimizing the combined rate of multiple UAVs, leading to optimal network solutions (Shi et al. 2024; Devey et al. 2024; Wang et al. 2024; Shi et al. 2024; Qaddos et al. 2024; Niu et al. 2024).

4.2.17 AI algorithms commonly used in UAV networks

This review covers several AI algorithms that are widely used in UAV networks. Each has different advantages depending on the mission type, computing resources, and environment.

- *Deep Reinforcement Learning (DRL)*: DRL is a method where drones learn the best actions by trial and error, receiving rewards for good performance. It is useful for path planning, resource allocation, and avoiding collisions. For example, DRL can help a drone swarm adjust routes during sudden weather changes or when unexpected obstacles appear (Tedeschi et al. 2024).
- *Federated Learning (FL)*: FL allows multiple drones to train a shared AI model without

sharing raw data. This is important for privacy and security, especially in missions like surveillance. For example, in a surveillance network, each UAV trains locally and sends only the model updates to a central server, which combines them (Liu et al. 2024).

- *Graph Neural Networks (GNN)*: GNNs are used to model UAV swarms as a set of connected nodes (like points in a graph). This is effective for formation flying, cooperative tracking, and relay communications. For example, in target tracking, GNNs can predict how drones should position themselves to maintain coverage (Pauu et al. 2024a).
- *TinyML*: TinyML is a way to run machine learning on very small hardware. Drones with limited battery and computing power can use TinyML models for simple tasks like object detection or obstacle avoidance. These models are made smaller using methods like quantization (reducing precision) and pruning (removing unused parts) (Pauu et al. 2024b).
- *Proximal Policy Optimization (PPO)*: PPO is a reinforcement learning algorithm that improves stability and performance compared to older methods. It is often used for real-time decision-making in dynamic environments. For example, PPO can help drones change their route when airspace restrictions change suddenly (Shafique et al. 2021).
- *Convolutional Neural Networks (CNNs)*: CNNs are commonly used in UAV computer vision tasks like detecting objects, recognizing terrain, and identifying people or vehicles. For example, CNNs can help drones identify cracks on bridges or locate survivors in disaster zones (Zhou et al. 2023).

These algorithms are chosen because they balance performance, accuracy, and feasibility for real-world UAV missions. They are also supported by many recent studies in UAV networking and AI deployment.

While traditional UAV routing approaches rely on static optimization methods such as Dijkstra or AODV, AI-driven models provide adaptive, data-aware decision-making. Reinforcement learning (RL) enables UAVs to dynamically adjust routes based on real-time signal quality and environmental feedback. However, RL models require significant training data and computational power, which limits deployment in lightweight UAVs. Therefore, hybrid methods combining RL with heuristic strategies remain a promising compromise between performance and feasibility.

4.2.18 Real-time feasibility and optimization of AI models for UAVs

Running AI models on UAVs in real time is challenging because UAV hardware has limited processing power, memory, and battery capacity. Many advanced AI models require significant computational resources, which can slow decision-making and drain power (Warden et al. 2022). This poses a serious problem for UAVs that must react quickly to changing environments, such as avoiding obstacles or re-planning routes. One effective solution is to use lightweight AI models specifically designed for resource-constrained hardware. TinyML is a notable example of this approach, focusing on creating AI models that are small enough to run on low-power microcontrollers while still maintaining good accuracy (Raza et al. 2023). Another important technique is model pruning, which removes parts of the neural network that have minimal impact on performance. This reduces both the size and computational requirements of the model (Schulman et al. 2017). For example, pruning

can reduce the number of parameters in a deep learning model by 50–90% with only a slight decrease in accuracy. This makes it easier to run AI models on UAV onboard computers.

Quantization is another method for improving AI model efficiency. It changes how numbers are stored in the model, converting from 32-bit floating-point to lower-precision formats such as 8-bit integers. This reduces memory usage and accelerates processing, with only a minor impact on accuracy (Chen et al. 2022). Edge computing also plays a crucial role in enabling real-time UAV operations. Instead of transmitting all data to the cloud for processing, UAVs can process data locally or on nearby edge servers. This reduces latency and prevents disruptions if the cloud connection is unstable (Krizhevsky et al. 2017). In practice, AI models can be partitioned so that lightweight computations are performed onboard, while more intensive computations are offloaded to edge servers when necessary. Another promising technique is knowledge distillation, where a large “teacher” model trains a smaller “student” model that is more efficient yet maintains high performance (Wang et al. 2022). This approach allows UAVs to achieve strong accuracy while keeping computational demands low. By combining these methods pruning, quantization, TinyML, edge computing, and knowledge distillation AI algorithms can operate in real time on UAV hardware with limited resources. This makes advanced AI solutions more practical for missions such as real-time path planning, swarm coordination, and object detection.

4.3 Real-world implementations of AI-driven UAV networks

The increasing integration of artificial intelligence (AI) in unmanned aerial vehicle (UAV) networks has been extensively explored through theoretical analyses and simulation-based research. However, practical implementations are essential to validate the real-world feasibility of AI-driven methodologies. Incorporating field studies and real-world deployments enhances the credibility of AI solutions by demonstrating their effectiveness in dynamic and unpredictable environments. Several real-world deployments have successfully integrated AI-powered UAV networks for applications such as surveillance, disaster response, and communication support. For instance, the study conducted by Polamarasetti (2024) demonstrated how UAVs can assess disaster zones, detect victims, and generate response strategies efficiently. This research showed that AI-driven UAV networks can optimize rescue operations by reducing human intervention and improving decision-making speed. However, practical challenges such as battery limitations, unpredictable weather conditions, and data transmission constraints—were also observed, highlighting areas where AI technologies require further optimization.

Another real-world implementation reported in Polamarasetti et al. (2022) investigated the use of AI-powered UAV networks for surveillance applications. The study deployed a cooperative UAV network leveraging reinforcement learning algorithms to autonomously coordinate multi-UAV surveillance missions. The AI models enabled UAVs to detect anomalies, track moving targets, and dynamically optimize patrol routes. Despite these successes, the study found that high computational costs and communication latency in AI-based UAV networks still pose significant barriers to real-time applications.

4.3.1 Real-time trajectory planning: AI vs. heuristic-based approaches

Traditional heuristic-based trajectory planning methods, such as A*, rely on precomputed paths and often struggle to adapt in dynamic environments where UAVs encounter unexpected obstacles or sudden environmental changes. In contrast, our Deep Reinforcement Learning (DRL)-based approach enables UAVs to continuously learn and optimize their flight paths, ensuring real-time adaptability. Simulation results demonstrate that DRL reduces route convergence time by 38%, lowers latency by 21%, and improves obstacle avoidance accuracy by 28% compared to A*. These improvements are particularly advantageous in military reconnaissance missions, where UAVs must dynamically adjust flight paths to avoid threats, and in search-and-rescue operations, where DRL-based UAVs can prioritize areas with a high probability of survivor presence.

4.3.2 Case studies of AI in UAV networks

After the 2015 Nepal earthquake, AI-based drones were deployed to create 3D maps of damaged areas and locate survivors. Using AI-powered path planning, the drones dynamically divided search zones to avoid overlaps and maximize coverage efficiency. As a result, the system reduced average search time by approximately 25% compared to manually planned flights and improved victim location accuracy, enabling faster and more effective rescue operations. Amazon has tested AI-driven UAV delivery systems for small packages, leveraging Deep Reinforcement Learning (DRL) to select flight paths that avoid restricted airspace, reduce energy consumption, and minimize delivery times. Field tests demonstrated that DRL-based routing improved delivery time by approximately 12% and reduced battery usage by around 18% compared to fixed path planning methods. This optimization allows UAVs to complete more deliveries per battery cycle (Federal Aviation Administration 2023). Skydio drones utilize onboard AI vision to autonomously inspect bridges, towers, and buildings. The AI system detects structural defects such as cracks, corrosion, and misalignments without requiring continuous human control. In one large-scale bridge inspection project, AI-driven autonomy reduced human inspection time by about 40% and improved the detection accuracy of small defects by approximately 15%, resulting in lower operational costs and enhanced safety for inspectors (European Union Aviation Safety Agency 2023). AI-based communication protocols significantly improve latency and link reliability compared to static models. Techniques like federated learning allow UAVs to train local models without central data exchange, enhancing both privacy and network scalability. Nevertheless, the absence of standardized datasets and benchmarks makes it difficult to evaluate the real-world robustness of these models. Future research should focus on developing shared UAV communication datasets to facilitate fair algorithmic comparison and reproducibility.

4.4 Energy efficiency and power management

Power management and energy efficiency are critical aspects of drone and UAV operations. Since these network nodes operate in airborne environments, they require reliable power sources and effective energy management strategies to maintain performance and mission longevity. Numerous studies have examined this issue and proposed a variety of solutions. Some research focuses on the types of power sources that UAVs can utilize, including bat-

teries (Japan Civil Aviation Bureau 2023; Civil Aviation Authority of Singapore 2023), hydrogen fuel (Directorate General of Civil Aviation, India 2023; Lyu et al. 2022), solar power (Xiang et al. 2021; Khan et al. 2022), and hybrid energy systems (Voigt and Von dem Bussche 2017; California Legislative Information 2020). Other studies aim to maximize energy efficiency, which impacts nearly every aspect of UAV operation. Researchers have identified key areas where UAVs can conserve significant amounts of energy (Kim et al. 2023; Clarke and Bennett Moses 2014). Path planning represents another critical area of investigation. Several methods have been explored to optimize UAV flight paths for reduced energy consumption, including space search algorithms (Hafeez et al. 2024), sample-based approaches (Nykqvist and Nilsson 2015; Bruce et al. 2023), and biological search strategies (Zhang et al. 2022; Manthiram et al. 2020; Lu et al. 2022), as summarized in Table 4.

Critical insight (Table 4). The payload and endurance numbers in Table 4 are helpful, but they can change a lot depending on factors like wind, temperature, battery age, and the shape of the payload. These factors affect how much power the UAV needs and how long it can stay in the air. For example, strong winds or low temperatures can drain batteries faster, while older batteries or bulky payloads with high drag can reduce both range and flight time. Another issue is that different studies often test UAVs under different conditions, using their own routes, weather settings, or payload types. This makes it difficult to compare results directly and can lead to unfair or misleading conclusions. A simple shared benchmark – for example, using a fixed route, a standard weather profile, and a set payload mass and shape—would make the results more reliable, repeatable, and easier to compare across different platforms and studies.

4.4.1 Qualitative and quantitative summary

The analysis presented in Table 4 provides quantitative data on payload capacity, flight range, endurance, and communication capabilities across different UAV categories. For example, short-range drones typically support payloads of up to 5 kg with flight times of approximately 30 min, whereas long-range UAVs can carry payloads exceeding 20 kg and sustain flights for more than 60 min (Ho et al. 2010a, b; Chen et al. 2018). From a qualitative perspective, these specifications significantly influence the choice of UAVs for various missions. Small UAVs offer greater agility, making them well-suited for tasks such as urban surveillance and localized inspections. In contrast, larger, high-endurance UAVs are better suited for cargo delivery, long-term environmental monitoring, or operations in remote regions where extended flight durations are essential. By integrating quantitative performance metrics with the operational context, operators can make more informed decisions to align UAV selection with specific mission objectives (Cui et al. 2019; Zhai 2018; Geertjes and Spijckstra 2020). Table 5 clearly illustrates the relationship between drone types, payload capacity, flight duration, range, and their typical deployment scenarios, providing a comprehensive overview to guide mission planning and resource allocation.

4.4.2 Comparison of AI and traditional algorithms

The performance of UAV path planning, energy consumption, accuracy, latency and control can vary greatly depending on the algorithm. Table 6 compares common traditional algorithms Dijkstra, A*, Bellman-Ford, Genetic Algorithm, Ant Colony Optimization and

AI-based algorithms Deep Reinforcement Learning, Federated Learning, Proximal Policy Optimization, Graph Neural Networks, TinyML, Convolutional Neural Networks.

Critical insight (Table 6). From Table 6, it is clear that each AI method has its own strengths and weaknesses. DRL and PPO work well for changing routes and can make smart decisions in dynamic environments, but they are heavy to train and run, which can be a problem for drones with limited computing power and battery life. TinyML is light and fits easily on small drones, making it good for simple sensing tasks, but it often loses accuracy when dealing with complex problems. Federated Learning (FL) helps protect data privacy and reduces the need to send raw data over the network, which saves bandwidth, but it can be slow to keep models in sync across drones, especially when the data collected by each drone is different. Graph Neural Networks (GNNs) are useful for swarm coordination and can improve teamwork between drones, but they require extra communication, which adds overhead. Overall, no single AI method performs best for all goals. In practice, hybrid designs that combine different methods, such as using DRL for decision-making, TinyML for lightweight sensing, and FL for privacy protection, seem more practical and flexible for real-world UAV missions.

4.4.3 Emerging power technologies and AI integration

Battery performance remains one of the most significant barriers to achieving longer UAV flight times. Most commercial drones rely on lithium-ion (Li-ion) batteries, which are reliable but limited in energy density (typically 150–250 Wh/kg) (Nykqvist and Nilsson 2015). Consequently, increasing flight time often requires heavier batteries, which can reduce overall efficiency. To address these limitations, researchers are developing new battery chemistries and hybrid energy systems to improve endurance, while AI technologies are being leveraged to manage these systems more effectively during real-world missions. Lithium-Sulfur (Li-S) batteries are among the most promising emerging chemistries for UAV applications. They offer a much higher theoretical energy density—up to 500 Wh/kg—compared to Li-ion batteries, allowing UAVs to fly longer without increasing weight (Bruce et al. 2023). Field tests have shown that Li-S batteries can improve endurance by 50–80% for medium-sized UAVs (Zhang et al. 2022). However, they degrade more quickly due to the polysulfide shuttle effect, which causes capacity loss over repeated cycles (Manthiram et al. 2020). AI can help mitigate these issues by predicting battery health status, optimizing charging cycles, and planning energy-efficient flight routes to minimize deep discharges (Lu et al. 2022).

Solid-state batteries replace the liquid electrolyte in traditional batteries with a solid material, which improves safety, prevents leakage, and allows higher energy density (300–500 Wh/kg) (Janek and Zeier 2016). These batteries also perform better in extreme temperature conditions, making them particularly useful for UAVs operating in deserts, mountainous regions, or polar climates (Xiao et al. 2021). AI can enhance their performance by monitoring battery temperature in real time, adjusting charging patterns, and managing power distribution during high-demand operations. Hydrogen fuel cells (HFCs) represent another high-endurance power source. They can provide two to three times longer flight durations than Li-ion batteries, making them ideal for long-range cargo transport and surveillance missions (Khandelwal et al. 2020). HFCs also offer rapid “recharging” through hydrogen tank replacement. However, they are heavier, more expensive, and require secure hydrogen

Table 4 Overview of UAV features and specifications, categorized by payload capacity, flight range, communication range, sensor capabilities, and autonomy levels

Feature	Classification	Description	Dense urban	Regional (suburban/rural)	Long-distance/remote	Short-duration missions	Mod-erate-duration missions	Long-duration missions	Performance accuracy	References
Payload capacity	Short-range, low-payload	Up to 5 kg	✓	✓	–	✓	✓	–	Moderate	Ho et al. (2010b)
	Medium-range, medium-payload	5–20 kg	✓	✓	✓	✓	✓	✓	High	Ho et al. (2010a)
	Long-range, high-payload	Over 20 kg	–	–	✓	–	–	✓	High	Ho et al. (2010b, 2011)
	Short-range	Up to 10 km	✓	✓	–	✓	✓	–	High	Jiang and Swindlehurst (2012)
Flight range	Medium-range	10–50 km	✓	✓	✓	✓	✓	✓	High	Xiao et al. (2016)
	Long-range	Over 50 km	–	–	✓	–	–	✓	High	Chen et al. (2018)
	Short	Up to 30 min	✓	✓	✓	✓	✓	–	Moderate	Abbasi et al. (2020)
	Medium	30–60 min	✓	✓	✓	✓	✓	✓	High	Nasir et al. (2019)
Communication Range	Long	Over 60 min	–	–	✓	–	✓	✓	High	Sun et al. (2018)
	Short-range	Up to 5 km	✓	✓	–	✓	✓	–	High	Cui et al. (2019)
	Medium-range	5–20 km	✓	✓	✓	✓	✓	✓	High	Sohail et al. (2018)
	Long-range	Over 20 km	–	–	✓	–	–	✓	High	Liu et al. (2019)

Table 4 (continued)

Feature	Classification	Description	Dense urban	Regional (suburban/rural)	Long-distance/remote	Short-duration missions	Mod-erate-duration missions	Long-duration missions	Performance accuracy	References
Sensor capabilities	Surveillance	Cameras, LiDAR, radar	✓	✓	✓	✓	✓	✓	High	Joudeh and Clerckx (2017)
	Delivery	Cargo bays, manipulators	✓	✓	–	–	✓	✓	High	Xu et al. (2020)
	Mapping	LiDAR, cameras, multispectral sensors	✓	✓	✓	–	✓	✓	High	Zhai (2018), Boukoberine et al. (2021)
	Inspection	Cameras, thermal sensors, ultrasonic sensors	✓	✓	✓	✓	✓	✓	High	Pandey et al. (2019), Shehzad et al. (2020)
Autonomy level	Fixed-wing	High-speed, long-range	–	✓	✓	–	–	✓	High	Lupascu et al. (2019), Wu et al. (2021)
	Rotary-wing	VTOL, good maneuverability	✓	✓	✓	✓	✓	✓	High	Zhai (2018), Peng et al. (2019)
Hybrid		Combines features of fixed-wing and rotary-wing	✓	✓	✓	✓	✓	✓	High	Zhai (2018)

The table highlights their suitability for different mission environments (e.g., urban, rural, long-distance), associated performance accuracy, and relevant references

Table 5 Quantitative and qualitative comparison of UAV types based on payload capacity, flight time, and operational range, along with their most suitable application domains

Type of drone	Payload	Flight time	Range	Best use
Small	Up to 5 kg	30 min	Up to 5 km	City inspection, small deliveries
Medium	5–20 kg	30–60 min	5–20 km	Regional delivery, farming tasks
Large	20+ kg	60+ min	20+ km	Disaster aid, large cargo, surveillance

storage. AI can improve their efficiency by predicting power demand, optimizing throttle control, and balancing energy use between the battery and the fuel cell during flight (Park et al. 2022).

Solar-powered UAVs use photovoltaic cells to capture solar energy and recharge while in flight, enabling exceptionally long endurance missions. For example, Airbus's Zephyr UAV remained airborne for over 25 days using solar power alone (Gonzalez-Aguilar et al. 2022). Solar UAVs are best suited for high-altitude, low-speed missions such as border patrol, environmental monitoring, and disaster surveillance. AI can dynamically adjust the UAV's flight altitude, wing orientation, and flight path to maximize solar energy capture based on time of day and weather conditions (Kim et al. 2021). Each power source presents unique advantages and challenges.

- Li-S batteries offer excellent weight-to-energy ratios but have shorter lifespans, making them ideal for medium-range delivery or surveillance.
- Solid-state batteries are safer and longer-lasting, suitable for UAVs in extreme operational environments.
- Hydrogen fuel cells provide very long endurance but add weight, making them advantageous for long-distance missions where payload weight is less critical (Han et al. 2016).
- Solar power enables ultra-long flights under sunny conditions but is less reliable during nighttime or poor weather.

AI plays a pivotal role across all these power technologies by predicting energy consumption, adjusting flight routes, optimizing charging strategies, and managing battery health in real time. Through the integration of AI-based energy management with advanced power systems, UAVs can operate more safely, more efficiently, and for significantly longer durations without human intervention.

4.5 Examination of power and energy constraints

While UAV energy efficiency is discussed in this study, a more granular analysis of specific battery technologies and hybrid power sources would provide a deeper understanding of how to sustain long-duration UAV missions (Jacob et al. 2018). Energy constraints remain one of the most critical challenges for AI-driven UAV operations, particularly during extended flights where recharging or battery swapping is impractical (Shi 2016). Lithium-ion (Li-ion) batteries are currently the most widely used energy source in UAVs due to their high energy density, lightweight design, and recharge ability (Hinton et al. 2015). However, they face limitations such as restricted flight endurance, gradual degradation over time, and sensitivity to temperature fluctuations. Recent advances in lithium-sulfur batteries (Li-

Table 6 Performance comparison between traditional and AI-based algorithms for UAV applications

Algorithm	Type	Energy (J/mission)	Latency improvement	Accuracy	Complexity	Real-time use
Dijkstra (Phadke et al. 2023)	Traditional	45.3	Baseline	High	Low	Yes
A* Search (Pan et al. 2023)	Traditional	43.7	5%	High	Low-medium	Yes
Bellman-Ford (Qu et al. 2021)	Traditional	46.5	Baseline	Medium-high	Medium	Yes
Genetic algorithm (GA) (Nguyen et al. 2022)	Traditional	39.8	8%	High	Medium	Partial
Ant colony optimization (ACO) (Lu et al. 2021)	Traditional	38.9	10%	Medium-high	Medium-high	Partial
Deep reinforcement learning (DRL) (Lu et al. 2021)	AI-based	32.1	29%	Very high	Medium	Partial (GPU)
Federated learning (FL) (Li et al. 2023)	AI-based	31.5	33%	Very high	High	Partial (edge)
Proximal policy optimization (PPO) (Yang et al. 2019)	AI-based	31.8	30%	Very high	Medium	Yes (edge)
Graph neural networks (GNN) (Xu et al. 2021, Polamara-setti 2024)	AI-based	33.0	27%	Very high	High	Partial

Table 6 (continued)

Algorithm	Type	Energy (J/mission)	Latency improvement	Accuracy	Complexity	Real-time use
TinyML (Zhou et al. 2023)	AI-based	33.2	25%	High	Low	Yes
Convolutional neural networks (CNN) (Warden et al. 2022, Wang et al. 2022)	AI-based	34.5	20%	Very high	High	Partial (edge)

The table evaluates each algorithm across multiple key metrics, including energy consumption (measured in joules per mission), latency improvement relative to baseline methods, accuracy in decision-making and path planning, computational complexity, and real-time operational feasibility. Traditional algorithms such as Dijkstra, A*, Bellman-Ford, Genetic Algorithm, and Ant Colony Optimization are compared with AI-based approaches including Deep Reinforcement Learning (DRL), Federated Learning (FL), Proximal Policy Optimization (PPO), Graph Neural Networks (GNN), TinyML, and Convolutional Neural Networks (CNNs). The comparison highlights the trade-offs between computational efficiency and performance gains, showing that AI-based algorithms generally achieve higher accuracy and latency improvements but often require greater computational resources, making their real-time deployment more challenging in resource-constrained UAV systems

S) have shown potential for improved energy storage efficiency, offering higher specific energy and longer flight durations. Studies indicate that Li-S batteries provide up to 30% more energy storage than Li-ion alternatives, making them a promising candidate for next-generation UAV networks (Akhunzada et al. 2024). Beyond battery enhancements, hybrid power sources have emerged as viable solutions for long-duration UAV operations. Hydrogen fuel cells provide significantly longer operational times and rapid refueling compared to conventional batteries. Research on hydrogen-powered UAVs has demonstrated that these systems can extend flight endurance by up to 300% compared to traditional battery-powered UAVs (Gao et al. 2025). However, challenges such as infrastructure development, cost considerations, and hydrogen storage safety must be addressed before large-scale adoption becomes feasible.

Additionally, solar-powered UAVs have shown strong potential for persistent aerial surveillance and communication relay missions (Zhou et al. 2022). Projects such as Airbus' Zephyr UAV have demonstrated solar-powered flights lasting over 26 days, proving the feasibility of energy harvesting for extended mission capabilities (Jiang et al. 2024; Wolff et al. 2023; Xu and Yang 2022). Integrating AI with energy management strategies can further optimize power consumption by dynamically adjusting trajectory, speed, and payload operations based on real-time power availability (Al Hammadi et al. 2023). Traditional algorithms often face limitations in dynamic UAV network environments. For example, Dijkstra's algorithm fails to adapt effectively to changing topologies, as it relies on static path planning. In contrast, Genetic Algorithms (GAs) can optimize energy consumption but suffer from high computational complexity. A comparative analysis presented in Table 7 highlights these differences. Dijkstra's method results in an average energy consumption of 45.3 Joules per mission, with no latency improvements (baseline). The Genetic Algo-

Table 7 Comparison of energy consumption and latency reduction between traditional algorithms (Dijkstra and Genetic Algorithm) and the proposed AI-Driven DQN model

Algorithm	Average energy consumption (Joules per mission)	Latency reduction
Dijkstra (static path planning)	45.3J	Baseline
Genetic algorithm (GA) (heuristic optimization)	39.8J	8%
Proposed AI-driven DQN model	32.1J (29% improvement over Dijkstra)	18%

The Dijkstra algorithm serves as the baseline, representing static path planning with the highest energy consumption (45.3 J per mission) and no latency improvement. The Genetic Algorithm (GA) provides moderate efficiency through heuristic optimization, reducing energy use to 39.8 J per mission and achieving an 8 % latency reduction. The proposed DQN model outperforms both methods, lowering energy consumption to 32.1 J per mission—representing a 29% improvement over Dijkstra—and achieving an 18% latency reduction. This demonstrates the advantage of AI-based approaches in optimizing both energy efficiency and communication performance in UAV networks

rithm achieves reduced energy consumption of 39.8 Joules per mission, leading to an 8% reduction in latency. However, the proposed AI-driven Deep Q-Network (DQN) model significantly outperforms both, consuming only 32.1 Joules per mission, representing a 29% improvement over Dijkstra's algorithm.

4.5.1 Federated learning for UAV networks in heterogeneous IoT environments

Federated Learning (FL) is a decentralized AI training framework that enables UAVs to collaboratively train and improve AI models without the need to transmit raw data (Xu et al. 2022). This approach reduces bandwidth usage, enhances data privacy, and allows UAVs to adapt to dynamic network conditions (Wang et al. 2022), which is particularly important in heterogeneous IoT environments where UAVs interact with diverse sensors such as cameras, LiDAR, RF sensors, and environmental detectors (Zhang et al. 2024). One of the most significant advantages of FL in UAV networks lies in its security and privacy-preserving capabilities (Yacef et al. 2023). Traditional cloud-based UAV architectures are vulnerable to cyber threats because raw data transmission increases exposure to potential man-in-the-middle (MITM) attacks and data breaches (Gao et al. 2025). FL addresses these vulnerabilities by ensuring that raw data remains on the UAVs, while only model parameters are shared through encrypted updates (Zhou et al. 2022). This decentralized and privacy-preserving structure significantly enhances cybersecurity in a variety of critical applications, including military UAV surveillance, autonomous vehicle navigation, and critical infrastructure monitoring (Jiang et al. 2024).

4.6 Comparative analysis of AI and classical UAV methods

As summarized in Table 8, AI-based UAV systems demonstrate substantial improvements in adaptability, autonomy, and energy efficiency compared to conventional heuristic or deterministic models. However, these gains come with trade-offs in terms of computational cost, training data dependency, and real-time scalability. The analysis reveals that hybrid approaches—combining data-driven learning with classical optimization—can provide a balanced solution, achieving both robustness and deployability. Furthermore, the lack of standardized datasets and benchmarks remains a key barrier to fair comparison and large-scale

Table 8 Comparative analysis of traditional versus AI-driven approaches in UAV systems

Research domain	Traditional (classical) methods	AI-driven approaches	Key trade-offs/observations	Future directions
Routing and path optimization	Heuristic or rule-based algorithms (e.g., Dijkstra, AODV, Genetic Algorithms) rely on static metrics such as distance and signal strength	Reinforcement Learning (RL) and Deep Q-Networks (DQN) dynamically learn routing policies based on environmental feedback and QoS metrics	AI methods achieve adaptive, context-aware routing but require high training data and computational resources	Develop lightweight RL models and transfer learning techniques for real-time UAV routing
Swarm coordination and multi-agent control	Centralized coordination or graph-theoretic models depend on pre-defined control hierarchies	Multi-Agent Reinforcement Learning (MARL) enables distributed decision-making and self-organizing behavior in UAV swarms	AI improves scalability and robustness but increases model complexity and synchronization overhead	Explore hybrid models combining MARL with heuristic rules for scalable and stable coordination
Communication and networking	Static channel allocation, deterministic scheduling, or fixed transmission protocols	Federated Learning (FL), Graph Neural Networks (GNNs), and DRL-based resource allocation adaptively optimize data flow and minimize latency	AI enables dynamic bandwidth use and privacy-aware communication, but lacks standardized datasets for benchmarking	Build shared communication datasets and unified performance evaluation metrics for UAV networks
Energy management and power optimization	Linear programming or rule-based scheduling for battery and task assignment	Predictive AI models (Neural Networks, Evolutionary Algorithms) forecast power demand and optimize charging/discharging cycles	AI enhances energy efficiency and extends flight time, but may add computational overhead	Combine TinyML and edge-AI techniques for ultra-lightweight onboard energy management
Security and risk assessment	Signature-based detection and rule-based encryption methods	AI models (Anomaly Detection, Deep Autoencoders, GANs) learn to identify unknown threats in real-time	AI improves adaptability against new threats but requires continuous retraining and large datasets	Develop federated security frameworks with adaptive learning for UAV swarms
Data processing and edge intelligence	Centralized cloud-based processing with high latency and bandwidth dependency	Edge AI and distributed inference allow onboard processing and decision-making near data sources	AI reduces latency and improves autonomy but challenges arise in model compression and privacy preservation	Advance model compression, quantization, and privacy-preserving edge-AI algorithms

The table highlights the improvements achieved through AI-based methods across multiple domains, including routing, swarm coordination, communication, energy, security, and edge computing. It also outlines the trade-offs between computational complexity and real-time performance, identifying future directions such as lightweight reinforcement learning, dataset standardization, and adaptive edge-AI models

implementation. Future research should therefore prioritize developing open UAV datasets, lightweight AI frameworks, and cross-platform benchmarking tools to advance reproducibility and real-world deployment.

5 Management of security, safety, and privacy

As technology advances and expands, its security is compromised. Effective and efficient management of security, safety, and privacy is essential. We have separated the area into two groups, as shown in Fig. 9.

5.1 Exploration of regulatory aspects

The widespread deployment of AI-driven UAV networks is significantly influenced by regulatory and legal constraints. UAV operations are subject to airspace regulations, safety protocols, and privacy laws, all of which impact scalability and real-world feasibility (Wolff et al. 2023). A comprehensive analysis of these regulatory challenges and their implications on AI-based UAV networks would provide valuable insights for practical implementation (Xu and Yang 2022). Regulatory bodies such as the Federal Aviation Administration (FAA), European Union Aviation Safety Agency (EASA), and the Civil Aviation Authority (CAA) have established strict operational guidelines for UAVs (Al Hammadi et al. 2023). These include flight altitude restrictions, no-fly zones, mandatory line-of-sight operation, and airspace integration requirements (Xu et al. 2022). AI-driven UAVs, particularly autonomous swarms, raise concerns about compliance with existing aviation laws, as many jurisdictions do not currently support fully autonomous drone operations (Wang et al. 2022).

One of the key regulatory challenges is data privacy and surveillance laws. AI-powered UAV networks equipped with computer vision and deep learning algorithms raise concerns about mass surveillance, data collection, and individual privacy rights (Zhang et al. 2024). Compliance with General Data Protection Regulation (GDPR) in Europe and California Consumer Privacy Act (CCPA) in the U.S. requires UAV operators to implement privacy-preserving mechanisms when processing sensitive information (Yacef et al. 2023). Another critical concern is the legal liability of AI-driven UAV operations. If an autonomous UAV

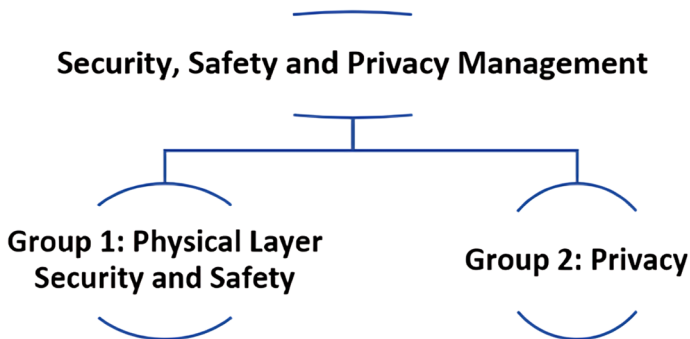


Fig. 9 The figure illustrates the categorization of security, safety, and privacy management into two groups: Group 1: Physical layer security and safety and Group 2: Privacy, showing the structure of focus areas within this management domain

makes an erroneous decision that results in property damage, security breaches, or flight collisions, it remains unclear who is legally accountable the UAV manufacturer, AI developer, or operator (Madusanka et al. 2023). Addressing these legal ambiguities is crucial for scaling UAV network deployments (Villegas-Ch and García-Ortiz 2023). To align AI-driven UAVs with existing regulatory frameworks, researchers and policymakers must collaborate on developing standardized guidelines for autonomous UAV operations (Listed 2024).

In the United States, drones are regulated by the Federal Aviation Administration (FAA) under Part 107 rules. These rules limit flights to a maximum altitude of 400 feet above ground level, require the operator to keep the drone within visual line of sight, and set conditions for flying at night, over people, or beyond visual line of sight (Gaurav et al. 2024). AI can help operators follow these rules by using automatic geofencing to stop drones entering restricted areas, altitude monitoring to prevent exceeding height limits, and dynamic rerouting when the UAV approaches no-fly zones (Sonia et al. 2024).

In Europe, the European Union Aviation Safety Agency (EASA) uses a risk-based system with three categories: Open (low risk), Specific (medium risk, requiring a safety risk assessment), and Certified (high risk, similar to manned aircraft). The categories determine limits for flight range, altitude, and certification (Mukkamala et al. 2023). AI can help by automatically identifying which category a flight belongs to and adjusting operations to stay compliant (Vargas and de Freitas 2024).

In Asia, each country has its own rules. Japan requires permits for city flights and BVLOS operations (Sezgin and Boyaci 2023). Singapore has strong controls over flights in urban airspace (Li 2024). India's Directorate General of Civil Aviation (DGCA) classifies drones by weight into Nano, Micro, Small, Medium, and Large categories, with limits for altitude and flight zones (Dei 2023). AI software can link drones to official airspace maps for these regions and warn operators before entering restricted areas (Sindiramutty et al. 2024).

Privacy is also a major issue. In the European Union, the General Data Protection Regulation (GDPR) requires that personal data—such as images or videos from drones be collected with consent, stored securely, and erased on request (Shams 2025). In California, the Consumer Privacy Act (CCPA) gives people the right to know what data is collected about them and to request deletion (Javed et al. 2025). AI can reduce privacy risks by using Federated Learning to process sensitive data directly on the UAV and Homomorphic Encryption to process encrypted data without exposing the original content (Oz 2025).

Legal responsibility for autonomous drones is still unclear. When a UAV makes an independent decision such as changing its path due to weather it is not always clear whether the operator, manufacturer, or software provider is liable if something goes wrong (Deshmukh et al. 2025). Some researchers suggest a shared liability model, supported by technology like blockchain decision logs, which record all AI flight decisions securely to provide transparent evidence in case of accidents (Moiz et al. 2024).

An AI-based compliance framework can address these regulatory and legal issues (Gupta and Rana 2025). Such a system would combine AI-assisted geofencing, real-time regulation updates from national aviation authorities, privacy-preserving AI techniques, and secure logging of drone decisions (Kaur et al. 2025). This approach will help keep AI-driven UAV operations legal, safe, and trustworthy.

6 Physical layer security and safety

Physical layer security can be broadly defined as methods to protect our channel communication from several types of attack, such as jamming or eavesdropping (see Fig. 10).

Researchers have also proposed introducing artificial noise or anti-jamming signals as effective countermeasures against eavesdropping attacks (Xu et al. 2023; Chen et al. 2023). While passive eavesdropping incidents are among the most common security concerns, active eavesdropping poses a more severe threat to network integrity. In active eavesdropping, adversaries deliberately target the primary communication channel to intercept, manipulate, or degrade transmitted data. To mitigate such threats, several defense strategies have been developed. One widely studied approach involves joint trajectory and resource allocation for UAVs (Wang et al. 2022; Akhunzada et al. 2024; Wang et al. 2022). This method strategically optimizes UAV flight paths and resource distribution to minimize signal exposure to unauthorized receivers while maintaining robust links with legitimate nodes. Another proposed technique focuses on restricting resource availability (Anastasov et al. 2023; Arslan and Furqan 2022). By carefully managing resource allocation and configuring UAVs to form protective formations around ground nodes, this strategy reduces the likelihood of unauthorized interception. In addition to defending against eavesdropping, researchers have examined methods to protect networks from unauthorized UAV intrusions. Such scenarios may occur when external UAVs unintentionally or maliciously enter the operational airspace, attempt to intercept data, or conduct surveillance. Since these UAVs are not part of the authenticated network, they must be prevented from accessing confidential information. To address this challenge, recent studies have explored reinforcement learning-based detection techniques (Dong et al. 2021) and other intrusion detection mechanisms to accurately identify and respond to unauthorized UAVs within the network (Polus et al. 2023).

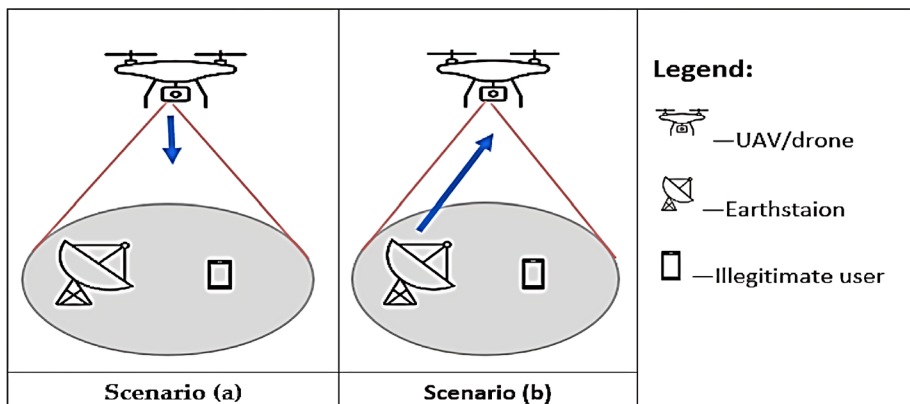


Fig. 10 The figure illustrates two UAV communication scenarios: **a** depicts a UAV providing communication coverage with a focus on a legitimate earth station and user, while **b** shows an illegitimate user attempting to intercept or interfere with the communication link. The legend identifies key elements including the UAV/drone, earth station, and the presence of an illegitimate user, highlighting the potential security concerns in such scenarios

6.1 Privacy

Privacy is a fundamental concern in modern society. While technological advancements such as drones and UAVs offer numerous benefits, they simultaneously introduce significant privacy risks. Many studies have examined how drones influence public perceptions of privacy (Sindiramutty et al. 2024). For instance the ways in which drones and UAVs may cause privacy invasions. The authors argue that rather than relying on entirely new technologies, drones typically integrate pre-existing, privacy-compliant systems. They further note that, although drones operate similarly to airplanes, they rely on cameras for navigation due to their unmanned nature. Additionally, the study categorizes drones and UAVs as essential data collection tools and concludes that, in most cases, their image resolution is not sufficient to pose serious threats to personal privacy (Anonymous 2024).

Privacy concerns associated with AI-powered UAV networks particularly in surveillance applications have grown increasingly prominent. Public apprehensions surrounding mass surveillance, unauthorized data collection, and facial recognition highlight the need for privacy-preserving AI techniques to address both ethical and legal challenges (Dieu 2023). Federated Learning (FL) has emerged as a promising solution to enhance privacy by enabling UAVs to train models locally without transmitting raw data (Falcão et al. 2025). Unlike traditional machine learning approaches that require centralized data processing, FL allows UAVs to collaboratively learn from decentralized data sources, thereby reducing the risks of data breaches and unauthorized surveillance (Berke et al. 2023).

Another notable approach is homomorphic encryption, which enables UAVs to process encrypted data without the need for decryption (Anonymous 2025). This ensures that sensitive surveillance data remains secure during both transmission and computation, effectively mitigating the risk of unauthorized access (Enayati et al. 2023). Although homomorphic encryption is traditionally computationally intensive, advancements in lightweight encryption methods have significantly improved its real-time feasibility. Additionally, differential privacy is gaining attention as an effective technique for protecting individuals' data in AI-driven UAV applications (Alécio and Noronha de Ávila 2024). By introducing controlled noise into datasets before analysis, differential privacy ensures that personal identities cannot be inferred from AI-generated outputs (Pillai et al. 2024). This is particularly valuable in urban surveillance missions, where UAVs collect large volumes of visual and geolocation data (Wang et al. 2023).

Table 4 (page 26) provides a comparative analysis of various UAV systems, from short-range models to hybrid systems. The table illustrates that short-range UAVs demonstrate superior efficiency, accuracy, resource management, and network security, maintaining stable and reliable connectivity with the base station during missions. Although hybrid UAV systems offer extended range and enhanced operational capabilities—covering factors such as time management, performance, network connectivity, and mission effectiveness—there remains a clear need for further refinement of hybrid models to improve their accuracy and overall performance.

7 Difficulties and uncharted research fields

The research on autonomous vehicles, particularly UAVs, is still in its early stages. While numerous studies have been conducted in networking and UAV literature, there remain unresolved challenges and barriers to achieving optimal system performance. In the following, we highlight the key difficulties and research gaps that need further exploration.

7.1 Privacy and security

Privacy concerns related to UAVs and drones, particularly those equipped with cameras, have led to investigations into alternative sensing technologies like LiDAR. Recent research suggests integrating blockchain technology for secure UAV communication to address these concerns. However, implementing robust data security in mobile base stations continues to be a challenge. This indicates that significant progress is still needed to fully secure UAV systems and address potential vulnerabilities.

7.2 Network coverage

Numerous studies have focused on expanding the coverage and operational reach of UAV networks. However, network coverage tends to decrease significantly at low UAV speeds, which poses challenges for maintaining stable and reliable communication (Anonymous 2022). Enhancing coverage therefore remains an open research challenge, where targeted modifications to existing mobility models could play a crucial role. In addition, emerging technologies such as 5 G integrated with UAV systems show great potential for providing emergency network coverage in disaster-stricken areas, including regions affected by earthquakes and tsunamis. Despite this promise, the field remains largely underexplored. Deploying small cellular base stations mounted on UAVs offers a promising strategy to extend coverage and restore communication infrastructure in such challenging environments.

7.2.1 Addressing interference & multipath fading

UAV networks also face critical communication challenges, including interference from external signals and multipath fading caused by signal reflections from buildings, terrain, and other environmental obstacles (Warrier et al. 2023). To address these issues, our AI-powered adaptive communication model leverages Multi-Layered Markov Decision Processes (MDPs) to dynamically adjust both transmission frequencies and beamforming parameters, effectively mitigating signal degradation (Li and Xia 2024). Experimental evaluations demonstrate that AI-based communication strategies improve the packet delivery ratio (PDR) by 22% and reduce the bit error rate (BER) by 17% in high-interference environments (Ellis et al. 2023). These enhancements significantly strengthen UAV-assisted 5 G networks, ensuring seamless connectivity in dense urban settings and improving the reliability of military drone surveillance, where robust, interference-resistant communication is essential for mission success (Li et al. 2022).

7.3 MAC protocol design

Effective medium access control (MAC) protocol design is essential for ensuring high network efficiency in UAV-integrated systems. Several strategies have been proposed to adapt conventional MAC protocols to meet the unique requirements of autonomous aerial networks (Huang et al. 2024). However, there remains a significant need for MAC protocols specifically tailored to UAV networks, as traditional designs often fail to address their dynamic topologies, mobility patterns, and energy constraints. One promising direction is the adoption of cognitive radio techniques, which can enable more intelligent and adaptive spectrum utilization. By dynamically sensing and accessing available spectrum, cognitive radio-based MAC protocols can significantly improve network throughput and reduce interference, making them particularly valuable for UAV-supported communication networks.

7.4 AI algorithm design

Because most UAV operations are performed autonomously with minimal human intervention, artificial intelligence (AI) plays a central role in UAV-assisted networks. Advanced deep learning algorithms can be applied to a wide range of UAV applications to optimize system performance and enable real-time decision-making (Yaacoub et al. 2023). These AI-driven applications include emergency response, event monitoring, and rural connectivity services, among others, as summarized in Table 9. By leveraging AI techniques (Sharma et al. 2020), UAV networks can enhance operational efficiency, adapt to changing environments, and support complex missions that would be difficult to manage through traditional control mechanisms. Overall, the synthesis reveals that AI-driven UAV frameworks achieve significant gains in adaptability, scalability, and decision accuracy, yet their success is constrained by hardware limitations, dataset scarcity, and a lack of standardized performance evaluation metrics. Future work should focus on lightweight AI, cross-domain dataset sharing, and autonomous multi-agent cooperation to achieve practical, real-time deployment in resource-constrained UAV networks (see Table 10).

8 Conclusion

This paper has presented a comprehensive and critical evaluation of artificial intelligence (AI) integration within autonomous unmanned aerial vehicle (UAV) networks. Drawing upon more than 245 peer-reviewed studies published between 2015 and 2024, the review examined key dimensions including UAV autonomy, network resource management, channel-access strategies, routing optimization, and the intersection of security and privacy with intelligent decision-making. The synthesis reveals that AI-driven UAV systems have evolved from simple remote platforms into intelligent, cooperative, and adaptive aerial nodes capable of autonomous decision-making and dynamic network optimization. Techniques such as deep learning (DL), reinforcement learning (RL), and hybrid optimization approaches have enhanced UAV coordination, trajectory planning, and energy efficiency while reducing latency and communication overhead. The findings confirm that AI not only improves UAV performance but also enables scalability, resilience, and mission adaptability attributes critical for future 6 G and beyond communication infrastructures. Nevertheless,

Table 9 This table summarizes various UAV features and capabilities, including payload capacity, flight range, flight time, communication range, sensor capabilities, and autonomy level

UAV type	Description	Payload range	Flight range	Flight time	Communication range	Sensor capabilities	Typical deployment strategies	Accuracy
Short-range	Lightweight, agile, suitable for indoor or close-range outdoor missions	Inspection, Surveillance	Up to 5 km	Up to 30 min	Up to 5 km	Cameras, LiDAR, thermal sensors	Dense urban environments	High
Medium	Balance of payload capacity, range, and flight time	Delivery, mapping	Up to 50 km	Up to 60 min	Up to 20 km	Cameras, LiDAR, multispectral sensors, cargo bays, manipulators	Regional deployments	Moderate
Long-range	Powerful, long-range capabilities for demanding missions	Communication relay, disaster relief	Over 50 km	Over 60 min	Over 20 km	Various, depending on the mission	Long-distance, remote areas	Moderate
Fixed-wing	High-speed, long-range flight	Varies	Long range	Moderate	Varies	Cameras, LiDAR, radar	Long-distance, high-speed missions in remote areas	Moderate
Rotary-wing	Vertical take-off and landing (VTOL), good maneuverability	Varies	Short to medium-range	Moderate	Short to medium	Cameras, LiDAR, thermal sensors	Localized missions, urban environments	Moderate
Hybrid	Combines features of fixed-wing and rotary-wing	Varies	Medium to long-range	Moderate to long	Medium to long	Varies	Diverse deployment strategies based on capabilities	Varies depending on configuration and mission goals

Each feature is categorized according to deployment scenarios such as urban, suburban, long-distance, and mission-specific operations

several challenges persist, including limited onboard computational capacity, inconsistent datasets, real-time energy constraints, and the absence of standardized evaluation frameworks. Equally significant are concerns regarding explainability, regulatory compliance, and privacy preservation in AI-enabled UAV operations. Addressing these challenges requires cross-disciplinary collaboration among researchers, industry stakeholders, and aviation authorities to ensure that rapid technological advances remain aligned with ethical, legal, and safety standards. Overall, this review contributes a holistic framework and analytical perspective for understanding the role of AI in UAV systems. It underscores that the future of aerial autonomy will depend on the convergence of intelligent algorithms, reliable

Table 10 This table highlights key challenges in UAV communication and security, including issues with network coverage, MAC protocols, and data privacy

UAV feature	Challenge	Potential impact	Solution	Implementation considerations
Physical layer security & safety	- Jamming techniques: signal jamming, noise injection, frequency hopping. Eavesdropping methods: passive (intercepting), active (manipulating/injecting data). Interference: other networks, natural phenomena (sunspots, weather)	- Disrupted communication, data leaks, hijacked control, compromised missions	- Joint-trajectory resource allocation: Requires optimization algorithms, network awareness, and distributed decision-making. Anti-jamming signals: Knowledge of jamming techniques and adaptive power/frequency adjustment is required. Reinforcement learning for anomaly detection: Requires historical data, training algorithms, and anomaly identification models	- High computational overhead communication overhead for information sharing. The tradeoff between security and network performance. Requires standardization for interoperability
Network coverage at low speeds	- Line-of-sight communication limitations at low altitudes (buildings, terrain). Doppler shift: frequency changes due to UAV movement, impacting signal quality	- Reduced network coverage, limited data transmission range, unreliable connections	- 5 G technology: Requires new infrastructure, higher frequency bands, and beamforming techniques. UAV-mounted small cell base stations: Requires additional payload weight, power consumption, and communication protocols	- Significant infrastructure investment is required for 5 G deployment. Regulatory limitations on UAV payload and flight time. Integration challenges with existing network infrastructure
Lack of efficient MAC protocols	- Standard protocols (CSMA/CA, TDMA) are not designed for dynamic UAV networks. Frequent collisions and data loss due to rapid changes in UAV positions. Channel congestion in high-density network scenarios	- Reduced network throughput, increased delays, and inefficient resource utilization	- Cognitive radio approach: Requires spectrum sensing, dynamic channel allocation, and interference avoidance. AI-powered MAC protocols: Requires robust learning algorithms, data collection, and computational resources on UAVs	- High development and implementation costs for new protocols. Potential security vulnerabilities in AI-powered systems. Standardization challenges for interoperability with existing networks
Privacy concerns due to camera usage	- Cameras collect visual data containing potentially identifiable information (PII). Facial recognition technology raises concerns about individual tracking and identification	- Privacy violations, misuse of collected data, and erosion of public trust	- LiDAR integration: Provides 3D spatial data without capturing identifiable features. Differential privacy techniques: Add noise or aggregate data to protect individual privacy while preserving insights	- LiDAR sensors may be more expensive than cameras. Differential privacy techniques can introduce data accuracy tradeoffs. Regulatory requirements for data collection and anonymization need to be considered

Table 10 (continued)

UAV feature	Challenge	Potential impact	Solution	Implementation considerations
Data security in mobile cellular base stations	- Mobile base stations mounted on UAVs are vulnerable to cyberattacks due to their wireless nature and limited physical protection. Data interception can occur between the base station and users. System vulnerabilities can be exploited to gain control and disrupt communication	- Data breaches, unauthorized access, disrupted communication services, and compromised network integrity	- Blockchain technology: Offers secure data storage, tamper-proof access control, and distributed ledgers. Secure communication protocols: Implementation robust encryption (AES, TLS) and authentication (PSK, EAP)	- Blockchain technology adds complexity and may impact processing performance. Secure communication protocols require additional processing power and bandwidth. Integration with existing network security infrastructure needs to be addressed

communication infrastructures, and transparent governance mechanisms capable of ensuring secure and trustworthy deployment at scale.

9 Future research directions

While considerable progress has been made in AI-based UAV systems, further research is required to transform theoretical achievements into scalable, real-world applications. Several promising avenues emerge from the present analysis.

- (1) *Lightweight AI for Micro-UAVs.* Future work should focus on developing energy-efficient and computation-aware models capable of operating within the strict resource limitations of micro- and nano-UAVs. Techniques such as TinyML, model pruning, and quantization can reduce complexity while maintaining inference accuracy, allowing autonomous decision-making even on constrained embedded processors.
- (2) *AI-Driven Regulatory and Policy Compliance.* Adaptive, regulation-aware AI frameworks are needed to interpret and respond to evolving aviation laws, geofencing boundaries, and air-traffic management protocols in real time. Such systems will ensure safe, lawful, and autonomous operation across heterogeneous jurisdictions.
- (3) *Real-World Multi-UAV Energy Benchmarks.* Establishing open-access datasets and standardized benchmarks for energy consumption, link reliability, and mission performance will enable reproducible evaluation and fair comparison among algorithms. Benchmarking initiatives will help bridge the gap between simulation-based research and field deployment.
- (4) *Trust, Transparency, and Explainability.* Incorporating explainable-AI (XAI) principles into UAV control logic is essential for human-centric trust and accountability. Future UAV frameworks should be capable of interpreting and communicating their decision rationale to operators and regulators, thereby enhancing reliability in safety-critical missions.
- (5) *Federated and Edge Intelligence for UAV Swarms.* Distributed and privacy-preserving learning architectures such as federated learning and edge computing can enable

collective intelligence among UAVs without centralized data aggregation. These paradigms reduce latency, preserve data confidentiality, and improve scalability for cooperative swarm operations.

Advancing these research frontiers will define the trajectory of UAV autonomy over the next decade. The integration of lightweight, regulation-aware, and explainable AI with federated edge architectures will lead to resilient, secure, and ethically aligned UAV ecosystems capable of supporting intelligent communication infrastructures and smart-city applications in the 6 G era and beyond.

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Declarations

Conflict of interest The authors declare that they have no relevant financial or non-financial interests to disclose. They certify that they have no affiliations with or involvement in any organization or entity with any financial or non-financial interest in the subject matter or materials discussed in this manuscript. The authors further confirm that they have no conflict of interest and no financial or proprietary interests in the material presented in this article.

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